SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2 - COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL 2-COMPARATIVE ANALYSIS**
Weightage:	**40%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	**Btech IT**	**Date:**	**26/08/2026**

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I G Devendra Nath Reddy G certify that this submission Is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Scope of Items(20%)	Comparison Depth(25%)	Use of Evidence(20%)	Critical Thinking(20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS

1. Comparison of L1, L2, and L3 Cache

Cache memory is a high-speed memory placed between the processor and main memory. Its purpose is to reduce the time required to access frequently used instructions and data.

Comparative Table

Feature	**L1 Cache**	**L2 Cache**	**L3 Cache**
Location	Inside/very close to CPU core	Close to CPU core	Usually shared among CPU cores
Latency	Lowest	Higher than L1	Higher than L2
Bandwidth	Very High	High	High, but lower effective speed than L1/L2
Capacity	Smallest	Medium	Largest
Cost per Bit	Highest	High	Lower than L1/L2
Sharing	Usually private to each core	Usually private or core-associated	Often shared by multiple cores
Main Purpose	Immediate data/instruction access	Backup for L1	Reduces expensive DRAM accesses

Exact latency, bandwidth, and size depend on processor architecture, so the comparison is relative rather than fixed.

Memory Access Hierarchy

```
    Fastest
       |
       ▼
  ┌──────────┐
  │ L1 Cache │
  └────┬─────┘
       |
       ▼
  ┌──────────┐
  │ L2 Cache │
  └────┬─────┘
       |
       ▼
  ┌──────────┐
  │ L3 Cache │
  └────┬─────┘
       |
       ▼
  ┌──────────┐
```

```
│  DRAM   │
└─────────┘
   │
   ▼
Slowest
```

Impact on Processor Performance

L1 Cache

L1 cache provides the fastest access to frequently required instructions and data.

Impact:

- Minimizes CPU waiting time.
- Supports high instruction execution speed.
- Improves performance of frequently repeated operations.

If required data is available in L1, the processor can continue execution with minimal delay.

L2 Cache

L2 acts as the second level when data is not found in L1.

Impact:

- Reduces the number of accesses to L3 and DRAM.
- Provides larger storage for frequently used data.
- Helps maintain processor throughput.

Although slower than L1, L2 is still significantly faster than accessing main memory.

L3 Cache

L3 is generally larger and can be shared among multiple processor cores.

Impact:

- Reduces traffic to DRAM.
- Improves multicore data sharing.
- Supports applications with large working data sets.
- Reduces the penalty caused by L1 and L2 cache misses.

Overall Analysis

```
L1 Hit → Very Low Latency → Maximum CPU Performance

L1 Miss
  ↓
L2 Hit → Low Latency → Good Performance

L2 Miss
  ↓
L3 Hit → Moderate Latency → Acceptable Performance

L3 Miss
  ↓
DRAM Access → High Latency → CPU May Stall
```

Therefore, an efficient cache hierarchy improves processor performance by increasing the **cache hit rate**, reducing memory access latency, and improving overall throughput.

2. Difference Between SRAM and DRAM

SRAM and DRAM are semiconductor memory technologies, but they are designed for different purposes.

- **SRAM** is mainly used in cache memory.
- **DRAM** is mainly used as main memory.

Comparative Table

Feature	**SRAM**	**DRAM**
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Storage Method	Uses stable storage cells	Stores data as electrical charge
Refresh Required	No	Yes
Latency	Very Low	Higher than SRAM
Bandwidth	Very High for small, close-to-processor structures	High, especially for large sequential transfers
Throughput	High for cache accesses	High for bulk data transfer
Capacity	Lower	Higher
Density	Lower	Higher
Cost per Bit	High	Lower
Power per Bit	Generally higher for large capacities	Lower cost per bit but requires refresh
Main Application	Cache memory	Main memory

SRAM in Cache Design

SRAM is used for L1, L2, and often L3 caches because it provides:

- Very low access latency
- Fast repeated access
- High performance for CPU workloads

Example:

CPU
│
▼
SRAM-based Cache
│
▼
DRAM Main Memory

The disadvantage is that SRAM requires more chip area per bit, making large SRAM memories expensive.

DRAM in Main Memory Design
DRAM provides:

- High storage capacity
- Lower cost per bit
- Suitable bandwidth for large workloads
- Efficient storage of programs and application data

However, DRAM has greater latency than SRAM.

Overall Analysis

SRAM is preferred when speed and low latency are the main requirements.
DRAM is preferred when high capacity and lower cost per bit are more important.

A balanced computer system therefore uses:

CPU
│
▼
SRAM Cache
Low Latency
│
▼
DRAM Main Memory
High Capacity
│
▼
Storage
Very High Capacity

This combination balances **speed, cost, capacity, and throughput**.

3. Virtual Memory, Paging, and Cache Memory Comparison

Virtual memory and cache memory both improve system memory management, but they perform different functions.

Basic Concept

Virtual Memory

Virtual memory allows a system to use secondary storage as an extension of physical memory.

Paging

Paging is the mechanism used to divide memory into fixed-size units:

- **Pages** in virtual memory
- **Frames** in physical memory

When a required page is not available in physical memory, a **page fault** occurs and the operating system must retrieve it from storage.

Cache Memory

Cache memory stores copies of frequently used data closer to the processor.

Comparative Table

Feature	Cache Memory	Virtual Memory	Paging
Main Purpose	Reduce CPU access time	Provide a larger logical memory space	Manage movement between virtual and physical memory
Location	Between CPU and DRAM	Uses DRAM with backing storage	OS/hardware memory-management mechanism
Access Unit	Cache line	Virtual address/page	Fixed-size page/frame
Typical Access Latency	Very Low	Depends on DRAM residency	Very low if translation/residency succeeds; extremely costly when a page fault requires storage
Throughput Impact	Usually improves throughput	Can improve capacity and multiprogramming	Can reduce throughput when page faults are frequent
Managed By	Mostly hardware	OS + hardware	OS + MMU
Failure Condition	Cache miss	Page not resident	Page fault

Impact on System Throughput

Cache memory generally improves throughput because the processor spends less time waiting for data.

Virtual memory improves system flexibility and allows larger programs and more processes to run, but excessive paging can significantly reduce throughput.

Conclusion

- **Cache memory optimizes speed.**
- **Virtual memory optimizes usable memory capacity.**
- **Paging manages virtual memory efficiently but frequent page faults reduce performance.**

4. NAND Flash vs DRAM

NAND Flash and DRAM are used at different levels of a memory hierarchy.

Comparative Table

Feature	NAND Flash	DRAM
Type	Non-volatile	Volatile
Data Retention	Retains data without power	Loses data without power
Access Latency	Higher	Much Lower
Read/Write Granularity	Block/page-oriented	Byte/word-oriented at the memory interface
Bandwidth	High at storage-interface level but with higher access overhead	High memory bandwidth and low access overhead

Feature	NAND Flash	DRAM
Throughput	Good for bulk storage operations	Better for active memory workloads
Capacity	Very High	High
Cost per Bit	Lower	Higher
Endurance	Limited program/erase endurance	Not limited by flash-style program/erase cycles
Main Use	SSDs and persistent storage	Main memory

NAND Flash

NAND Flash is suitable for:

- Permanent storage
- Operating systems
- Applications
- Large files
- Databases and game assets

Advantages:

- Non-volatile
- High density
- Low cost per bit
- Large storage capacity

Limitations:

- Higher latency than DRAM
- Writes and erases are more complex
- Not suitable as a direct replacement for high-speed CPU main memory

DRAM

DRAM is suitable for:

- Active programs
- Operating system data
- Running applications
- Databases
- Large working data sets

Advantages:

- Low latency compared with NAND
- High bandwidth
- Suitable for frequent read/write operations

Limitation:

- Data is lost when power is removed.

Hierarchical Memory Design

CPU
↓
L1/L2 Cache

|
▼
L3 Cache
|
▼
DRAM
Active Data
|
▼
NAND Flash
Persistent Storage

Analysis

DRAM should store **actively executing and frequently accessed data**, while NAND Flash should store **large, persistent, less frequently accessed data**.

Using NAND Flash directly for CPU memory would significantly increase access latency. Therefore, NAND and DRAM complement each other rather than directly competing.

5. MRAM vs RRAM Comparison

MRAM and RRAM are emerging non-volatile memory technologies that aim to combine some advantages of memory and storage.

Comparative Table

Feature	MRAM	RRAM
Full Form	Magnetoresistive RAM	Resistive RAM
Storage Principle	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access Time	Very Fast	Potentially very fast
Throughput	High	Potentially high, depending on implementation
Energy Efficiency	Very good, especially for standby	Potentially very energy-efficient
Data Retention	Strong	Good, technology-dependent
Endurance	Generally high	Can vary by material and device design
Density	Moderate	Potentially high
Main Strength	Fast, durable persistent memory	Dense, low-power memory and computing applications

MRAM

MRAM stores information using magnetic states.

Advantages

- Non-volatile
- Fast access
- High endurance
- Low standby power

- Suitable for embedded and persistent memory applications

Suitable Applications

- Automotive systems
- Industrial controllers
- Embedded devices
- Persistent cache
- Fast non-volatile memory

RRAM

RRAM stores information by changing the resistance of a material.

Advantages

- Non-volatile
- Potentially low energy operation
- High density potential
- Suitable for emerging AI hardware
- Can support in-memory computing approaches

Suitable Applications

- Edge AI
- Neuromorphic computing
- Low-power embedded systems
- Dense non-volatile memory
- AI accelerator architectures

Performance Comparison

Access Time

Faster / More Mature Predictable Performance

MRAM
|
▼
RRAM

RRAM can also achieve very fast switching,
but actual performance depends strongly on
the specific device technology and implementation.
MRAM is generally attractive when predictable, fast access and high endurance are required.
RRAM offers strong potential for fast switching and high density but may have greater technology-dependent variability.

Throughput

Both technologies can support high throughput, particularly when designed in parallel arrays.

- **MRAM:** strong for reliable embedded and persistent-memory workloads.
- **RRAM:** attractive for dense arrays and parallel computation, including some AI-oriented architectures.

Energy Efficiency

MRAM provides:

- Low standby power
- No refresh requirement
- Good energy efficiency for persistent memory

RRAM provides:

- Potentially very low switching energy
- No refresh requirement
- Potential advantages for compute-in-memory architectures

Final Summary Table

Comparison	Lower Latency	Higher Capacity	Higher Bulk Bandwidth Potential	Best Application
L1 vs L2 vs L3	L1	L3	Architecture-dependent	CPU performance optimization
SRAM vs DRAM	SRAM	DRAM	DRAM for large transfers	Cache and main memory
Cache vs Virtual Memory	Cache	Virtual memory provides larger logical space	Cache improves immediate execution throughput	Speed vs capacity management
NAND vs DRAM	DRAM	NAND Flash	DRAM for active memory workloads	Main memory and persistent storage
MRAM vs RRAM	Both can be fast; MRAM is more mature for predictable embedded access	RRAM has higher density potential	Both can scale with parallelism	Persistent memory and emerging AI systems